

# Predicting When a Pedestrian Will Cross

A plain-English summary of the whole project (for supervisor & co-authors)

**The project in 30 seconds.** We built a small AI model that watches a pedestrian for half a second of video and predicts whether they are about to step into the road. Along the way we discovered the original results were inflated by a hidden data problem (the model was secretly "watching" people who had *already* started crossing). We fixed it, rebuilt everything honestly, and then stress-tested the model eleven different ways. The honest result is actually strong: using just two cheap inputs, the model is as accurate as much heavier systems and runs in well under a millisecond.

## 1. What problem are we solving?

Self-driving and driver-assist systems need to know, *before it happens*, whether a pedestrian is going to cross. That is "intention prediction" — different from just tracking where someone is. We use the public **PIE dataset** (dash-cam videos from a car driving around a city, with every pedestrian labelled).

Our model is deliberately simple. For each pedestrian it looks at only **two things** over 16 video frames (about half a second):

- the **box around the person** (where they are and how it moves), and
- the **car's own speed** (from the vehicle's sensors).

It feeds these into a **BiLSTM** — a small neural network good at short time-sequences — which outputs a probability of crossing. Most competing methods use 3 to 7 inputs (body pose, optical flow, depth maps, scene segmentation, etc.); we use 2.

## 2. The turning point: a hidden data leak

The first version scored an impressive AUC of 0.931 (AUC is an accuracy-style score from 0.5 = guessing to 1.0 = perfect). It looked too good and converged suspiciously fast. We investigated and found the cause: **data leakage**.

The way the data was cut, **68% of crossing pedestrians were already mid-crossing** inside the half-second the model was shown. So the model wasn't *predicting* a future crossing — it was often just *detecting* a crossing already in progress. That is cheating, even if unintentional.

This is a subtle mistake that, importantly, **most published papers in this area never check for** — which makes catching and fixing it one of our paper's contributions.

## 3. The fix: a clean, honest protocol

We rebuilt the dataset so every half-second window **ends strictly before** the crossing begins (anchored on PIE's own "crossing point" marker, 1–2 seconds ahead). We verified **0% leakage** and the dataset grew from 1,389 to 4,906 examples.

Re-running honestly, the score barely changed: **AUC 0.932** (vs the old 0.931). So the fix was a *credibility* win, not a performance loss — the model genuinely predicts the future, and now we can prove it. **0.932 is our headline number.**

## 4. What we found (the headline results)

- **Accuracy with a confidence range:** AUC **0.932**, 95% confidence interval **[0.92–0.95]**. This is at the **top** of the published results for this task, despite using the **fewest inputs**.
- **The car's speed is the secret sauce:** remove it and the score collapses from 0.932 to 0.753. The ego-vehicle's speed carries most of the signal (the instrumented car tends to slow for people it expects to cross). We flag this honestly as a limitation.
- **It is extremely fast:** one prediction takes **0.575 milliseconds** — about 58× faster than a single video frame's time budget. The slow part of a live system is the person-*detector* (YOLO, 93% of the time), not our model (4.5%).
- **It survives the real, messy pipeline:** swapping perfect hand-drawn boxes for a real automatic detector barely changes the score (drop of just 0.009).

## 5. Everything we checked (the eleven reviewer issues)

A journal reviewer would attack a result like this from many angles, so we did the attacking ourselves first. Each row is one self-check, in plain English.

| # | What we checked                          | Plain-English result                                                         |
|---|------------------------------------------|------------------------------------------------------------------------------|
| 1 | Was there a data leak?                   | Yes — 68% of crossers seen mid-crossing. Found and proven.                   |
| 2 | Rebuild cleanly & retrain                | Leak removed (0%); honest score 0.932; speed is the key input.               |
| 3 | How do we compare to others?             | Top AUC of the standard table, with the fewest inputs.                       |
| 4 | Is the score statistically solid?        | Yes — confidence interval [0.92–0.95] from 10,000 resamples.                 |
| 5 | Is our test set just an easy one?        | No — tested on all 6 city-sets, score holds (0.928 average).                 |
| 6 | Do the settings (window/horizon) matter? | Window length: no. Predicting further ahead: yes, it gets harder (sensibly). |

|    |                                                                   |                                                                                  |
|----|-------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 7  | Is the model the right size?                                      | Yes — bigger/deeper gives no real gain. 128-unit, 2-layer justified.             |
| 8  | Did we tune settings properly?<br>( <i>supervisor-requested</i> ) | Ran a 36-option search; it confirmed our hand-picked settings were already best. |
| 9  | How fast is it, really?                                           | 0.575 ms per prediction; the system speed is set by the detector, not us.        |
| 10 | Does a real detector hurt it?                                     | Barely (drop 0.009). The weak link is tracking, not our prediction.              |
| 11 | Tidy up all the project documents                                 | Done — docs now match reality.                                                   |

## 6. The bottom line (what the paper will claim)

On a **leakage-free, standard PIE protocol**, a model using only **bounding-box motion + the car's speed** reaches **AUC 0.932 [0.92–0.95]** — matching or beating much heavier multi-input systems — and runs in **0.575 ms per prediction**. We back this with a confidence interval, all-sets cross-validation, multi-seed ablations, a documented settings search, and a test with a real detector in the loop. **Less input is enough; the bottleneck is perception, not prediction.**

## 7. The one limitation we state up front

Much of the accuracy comes from the **car's speed**, which partly reflects the human safety-driver's own anticipation (they ease off for people they expect to cross). It is a legitimate real-world signal, but it is not "pure vision." We disclose this clearly and suggest a follow-up robustness test as future work.

## 8. Where everything lives

All experiments, numbers, and figures are in the **journal\_prep/** folder, one sub-folder per issue (issue1\_leakage\_audit/ ... issue10\_gt\_vs\_detector/), each with its own README and result files. The paper-writing workspace and instructions are in **Journal\_writing/**.